



21115-200
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Department of Electrical Engineering
END-TERM EXAMINATION, DECEMBER 2022
B.Tech. III semester
Subject: EMEC-I

Max. Marks: 50

Subject Code: EE-212

Time: 3 Hours

Note: All Questions are compulsory. Attempt the questions in sequence.

Q.No.	Question	Marks	CO
1. a.	Deduce the relationship between line and phase voltage and current in a three phase star and delta connected circuit. Draw relevant phasor diagrams.	05	CO2
1.b.	A 440/254 volts, 3-phase, 4-wire system supplies an unbalanced load represented by the following impedances in ohms connected between R, Y and B phase respectively and the neutral: $16 + j 12$, $14 - j 21$ and 25 ohms. The phase sequence in RYB. Calculate the current in each conductor of the cable and the readings on each of the three wattmeters connected in each line and the neutral.	05	CO2
2. a.	Explain with necessary connection and phasor diagrams how three phase supply can be converted into a balanced two-phase supply. If the load is balanced on one side, show that it will be balanced on the other side.	05	CO2
2.b.	A load of 100 kVA is to be supplied at 460 Volts from 2300 V supply mains by an auto transformer. Determine the current and voltage rating for each of the two windings. What would be the kVA rating of the transformer, if it were used as a two winding transformer?	05	CO4
3. a.	A 600 kVA load at a power factor of 0.7 lagging is supplied by three single phase transformers of 250 kVA, 2200/220 V, connected in delta-delta connection. If one transformer is damaged and removed from the service, calculate for the V-V bank: (i) the kVA load supplied by each transformer (ii) Percentage of load supplied by each transformer (iii) total kVA rating of the transformer in V-V bank (iv) ratio of V-V bank to delta-delta bank transformer rating (v) percentage increase in load on each transformer when one transformer is removed.	05	CO4
3.b.	Two transformers of 20 kVA each with turns ratio respectively of 250:1000 and 250:1025 are connected in back-to-back test; the two primaries being fed from a 250 V supply and secondaries being connected in phase opposition. A booster transformer connected on primary side to the same 250 V supply is used to inject a voltage in the circuit of secondaries so as to circulate a current of 20 A. The core loss of each transformer are 350 W and each transformer has a reactance 2.5 times its resistance. Calculate the possible readings of the wattmeter connected to measure the input to the primaries.	05	CO5

4.a.	Explain the process of voltage build up in a dc shunt generator? Sketch the magnetization characteristics of a dc shunt generator. From the characteristics, give expressions for critical resistance and critical speed.	05	CO3
4.b.	A 220 V dc generator supplies 4 kW at a terminal voltage of 220 V, the armature resistance being 0.4 ohms. If the machine is now operated as a motor at the same terminal voltage with the same armature current, calculate the ratio of generator speed to motor speed. Assume that the flux/pole is made to increase by 10% as the operation is changed over from generator to motor.	05	CO3
5.a.	Explain armature reaction and its effect on dc machine. Sketch the distribution of flux in air gap due to (i) Main field mmf only (ii) Armature mmf only (iii) Both main field and armature mmf.	05	CO3
5.b.	A dc shunt motor runs at 1200 rpm on no load drawing 5 A from 220 V mains. Its armature and field resistances are 0.25 ohms and 110 ohms respectively. When loaded (at motor shaft), the motor draws 62 A from the mains. What would be its speed? Assume that the armature reaction demagnetizes the field to the extent of 5%. Also, calculate the internal torque developed at no load and on load. What is the motor shaft torque at load (this torque drives the mechanical load).	05	CO3